

PROFILE

C++ • Distributed Systems • Backend Infrastructure • Databases

Senior Software Engineer with 18+ years of experience designing and developing high-performance, distributed, and mission-critical software systems. Deep expertise in C/C++, distributed systems, concurrency, multithreading, transaction processing, relational databases, networking, and performance optimization.

Extensive experience building scalable backend and infrastructure systems at Amazon, Tableau, Microsoft, Bloomberg, MySpace, and technology startups. Strong systems-programming background spanning distributed database technology, high-throughput services, real-time applications, network programming, and cloud-based systems.

TECHNICAL SKILLS

Languages: C, C++, Java, Kotlin, C#, Python, Rust

Distributed & Backend Systems: Distributed systems, Paxos, distributed transactions, replication, sharding, concurrent programming, multithreading, multiprocessing, client/server architecture, N-tier systems, REST and web services

Databases: PostgreSQL, Oracle, SQL Server, MySQL, transaction processing, database programming, caching, query processing

Systems & Networking: Linux/UNIX, Windows, embedded systems, TCP/IP, network protocols, real-time programming

Cloud & Frameworks: AWS, Akka, Temporal, .NET, ASP.NET

Performance Engineering: Scalability, performance optimization, high-throughput systems, reliability, profiling and system refactoring

Machine Learning / Scientific Computing: OpenCV, TarsosDSP, Wolfram Mathematica

Development: Git, Visual Studio, Unix development tools

PROFESSIONAL EXPERIENCE

AMAZON — Seattle, WA

Senior Software Engineer | 2022–2026

- Led design of an Android-based application integrated with AWS backend services.
- Designed and developed AWS services supporting continuous monitoring and data collection.
- Led implementation of machine-learning models and libraries using C++, Java, TensorFlow, OpenCV, and TarsosDSP.
- Refactored application components and architecture and developed production features across application and backend systems.

TABLEAU SOFTWARE — Seattle, WA

Senior Software Engineer | 2017–2022

- Improved performance of the query-processing pipeline.
- Contributed to implementation of classification technology for error tracking and prediction.

- Migrated server-side components from Windows to Linux.
- Developed production software and mentored junior engineers.

MICROSOFT — Redmond, WA Senior Software Engineer | 2016–2017

- Designed and implemented a local caching subsystem.
- Refactored and modernized components of the existing codebase.
- Improved software build and deployment processes

BLOOMBERG — New York, NY Senior Software Engineer | 2012–2013

- Led design and implementation of real-time Excel functionality supporting high-frequency sell-side trading and data flows.
- Developed a high-performance C++ RTD server integrated with C#/ .NET Excel extensions.
- Designed data-flow architecture and algorithms for real-time financial applications.
- Collaborated across teams on systems delivering large datasets to customer trading models and executing orders in real time.
- Optimized and refactored performance-critical application components.

TRANSLATTICE — Santa Clara, CA Senior Software Engineer | 2010–2011

Distributed Database / PostgreSQL Engineering

- Implemented the Paxos consensus algorithm in C for distributed database operations built on the PostgreSQL engine.
- Designed and implemented distributed transactions and distributed table sharding for PostgreSQL.
- Designed and implemented multithreaded and multiprocess communication and replication infrastructure.
- Developed performance-critical database-engine functionality in C, with supporting infrastructure and tools in C++.
- Optimized and refactored core distributed database components for performance and scalability.

MYSAPCE.COM — Seattle, WA Senior Software Engineer / Architect | 2007–2010

- Led architecture and implementation of the MySpace Messaging platform integrating messaging, email, comments, and WebIM.
- Designed transactional message-passing infrastructure and REST-based backend services using C#, .NET, ASP.NET, and SQL Server.
- Refactored backend architecture and designed a transition toward a more scalable multi-tier system.

AMAZON.COM — Seattle, WA Senior Software Engineer III | 2004–2007

- Large-Scale Distributed C++ Infrastructure
- Led design of a next-generation high-performance distributed software deployment platform.
- Designed and developed distributed, multithreaded C++ systems managing, updating, and monitoring 30,000+ hosts.

- Built an architecture spanning multiple server farms, server applications, client applications and daemons, database persistence, and data-streaming subsystems.
- Designed solutions addressing scalability, concurrency, performance, availability, and operational reliability.
- Developed across the infrastructure stack, including Oracle databases, application servers, web servers, and end-host tools.

EDUCATION

Master of Applied Mathematics
University of Washington, Seattle, WA

Master of Engineering
Simon Fraser University, Vancouver, BC, Canada

B.Sc. Hon in Electrical Engineering
School of Electrical Engineering, **University of Sarajevo**